

Power BI Basics Assignment

Question 1: What is Power BI and why is it used in businesses?

Answer:

Power BI is a business intelligence and data visualization tool developed by Microsoft that allows organizations to connect to various data sources, transform raw data into meaningful insights, and present those insights through interactive dashboards and reports. It is part of the broader Microsoft ecosystem and integrates naturally with tools like Excel, Azure, and SQL Server, making it widely accessible across different types of organizations.

Core Components of Power BI

Power BI Desktop is the application used by data analysts and developers to build reports and dashboards. It is installed locally on a computer and is where data modelling, transformation, and visualization design take place.

Power BI Service is the cloud-based platform where completed reports are published, shared, and accessed by business users through a web browser. It enables collaboration and centralized access across an organization.

Power BI Mobile is the application that allows users to view and interact with dashboards on smartphones and tablets, ensuring decision-makers have access to live data regardless of their location.

Why Businesses Use Power BI

Turning Raw Data into Visual Insights Businesses generate enormous volumes of data every day from sales systems, marketing platforms, financial databases, and operational tools. Power BI connects directly to these sources and transforms rows of raw numbers into charts, graphs, KPI cards, and trend lines that are immediately understandable to non-technical business users. Instead of reading spreadsheets, a manager can glance at a dashboard and instantly understand how the business is performing.

Real-Time Decision Making Power BI supports live data connections, meaning dashboards can refresh automatically as new data arrives. This allows businesses to monitor daily sales performance, track inventory levels, or observe customer behavior in near real time and respond to changes quickly rather than waiting for end-of-month reports.

Single Source of Truth One of the most valuable aspects of Power BI in a business context is that it creates a centralized, standardized view of data that everyone across the organization can access. Instead of different departments working from different spreadsheets that may contradict each other, Power BI connects to a single cleaned and validated data source and presents consistent numbers to all users, eliminating confusion and conflicting reports.

Integration with Existing Microsoft Tools Since most businesses already use Microsoft products such as Excel, SharePoint, Teams, and Azure, Power BI fits naturally into existing workflows. Analysts familiar with Excel can transition to Power BI with relatively low effort,

and reports can be embedded directly into Teams channels or SharePoint pages, bringing insights to where employees already work.

Self-Service Analytics Power BI is designed to empower business users who are not data engineers or developers to build their own reports and explore data independently. A sales manager can drag and drop fields to create a custom regional performance chart without writing a single line of code. This reduces dependency on the IT or data team for every reporting request.

Cost Efficiency Compared to enterprise-grade BI tools, Power BI is relatively affordable, especially for organizations already subscribed to Microsoft 365. This makes it accessible to small and medium-sized businesses that need professional reporting capabilities without large software investment.

Question 2: Name and explain the three main components of Power BI.

Answer:

Power BI is not a single application but a suite of interconnected tools that together cover the complete journey of data from raw input to shared business insight. Each component serves a distinct purpose and targets a different stage in the analytics workflow.

Component 1: Power BI Desktop

Power BI Desktop is a free Windows-based application that is installed locally on a computer and serves as the primary development environment for building reports and data models. It is primarily used by data analysts, BI developers, and technically inclined business users who are responsible for designing and building analytical solutions.

Within Power BI Desktop, users connect to data sources ranging from Excel files and SQL databases to cloud services and APIs. The built-in Power Query Editor allows users to clean, reshape, and transform data before it is loaded into the model, applying steps such as removing nulls, changing data types, merging tables, and filtering rows. Once the data is prepared, users build a data model by defining relationships between tables, creating calculated columns, and writing DAX measures to express business logic such as year-to-date revenue or customer retention rate. The final step in Desktop is designing the visual report layout by arranging charts, tables, KPI cards, slicers, and maps onto report pages.

Power BI Desktop is where all the analytical and design work happens before anything is made available to the wider organization.

Component 2: Power BI Service

Power BI Service is the cloud-based platform hosted at app.powerbi.com where reports built in Power BI Desktop are published, shared, and consumed by business users across the organization. It is the collaboration and distribution layer of the Power BI ecosystem and is the component that most non-technical business users interact with on a daily basis.

Once a report is published from Desktop to the Service, it becomes accessible through a web browser without requiring any software installation. Business users can view dashboards,

interact with filters, drill down into charts, and explore data without needing to understand how the underlying model was built. The Service allows reports to be organized into workspaces, which function as shared folders where teams can collaborate on the same reports and datasets. It also supports scheduled data refresh, meaning the underlying data can be automatically updated at defined intervals so dashboards always reflect current information without manual intervention. Additionally, the Service enables role-based access control, allowing administrators to control who can view, edit, or share specific reports, ensuring sensitive business data is only accessible to authorized users.

Power BI Service transforms individually built reports into organization-wide business intelligence assets that are centrally managed and consistently accessible.

Component 3: Power BI Mobile

Power BI Mobile is the application available for iOS and Android devices that allows business users to access their Power BI dashboards and reports from smartphones and tablets. It is designed for executives, field teams, and decision-makers who need to stay connected to business performance data while away from their desks.

The mobile application is not a development tool. Users cannot build or edit reports within it. Its purpose is purely consumption, providing an optimized viewing experience for dashboards and reports that have already been published to the Power BI Service. Reports automatically adapt their layout for smaller screens, and certain visualizations are specifically designed for mobile-first viewing. The application supports real-time data refresh, meaning a sales manager checking revenue figures on a phone during a meeting is seeing the same live data as a colleague viewing the same dashboard on a desktop browser. It also supports alerts and notifications, allowing users to set threshold-based triggers such as receiving a push notification when daily sales fall below a defined target.

Power BI Mobile ensures that the insights built in Desktop and shared through the Service remain accessible and actionable regardless of where a business user is physically located.

Question 3: Explain the Power BI workflow.

Answer:

The Power BI workflow refers to the end-to-end process of moving data from its raw source all the way to a shared business insight that decision-makers can act upon. It is a structured sequence of stages where each step builds on the previous one, and skipping or poorly executing any stage affects the quality of the final output.

Stage 1: Connect to Data Sources

The workflow begins in Power BI Desktop where the user establishes a connection to one or more data sources. Power BI supports a wide range of connectors including Excel files, SQL databases, cloud platforms like Azure and Google Analytics, web APIs, SharePoint lists, and many others. The user selects the appropriate connector, provides the necessary credentials or file path, and loads the raw data into the Power Query Editor for further processing.

This stage determines the foundation of the entire report. If the wrong source is connected or the connection is unreliable, everything built on top of it is compromised.

Stage 2: Transform and Clean Data

Once data is connected, it enters the Power Query Editor where transformation and cleaning operations are applied. This stage is equivalent to the transformation step in an ETL pipeline. The analyst removes unnecessary columns, filters irrelevant rows, changes data types, handles null values, renames fields for clarity, merges or appends multiple tables, and applies any business-specific formatting rules.

The goal of this stage is to ensure that the data entering the model is clean, consistent, and structured correctly. Every transformation step is recorded as a sequential query step, making the process transparent and repeatable with each data refresh.

Stage 3: Build the Data Model

After transformation, the cleaned data is loaded into the data model. Here the analyst defines relationships between different tables, similar to how foreign keys work in a relational database. For example, a Sales table is connected to a Customers table through a shared Customer_ID field, and to a Products table through a shared Product_ID field.

This stage also involves creating calculated columns using DAX, which is Data Analysis Expressions, the formula language used in Power BI. Measures such as Total Revenue, Year-to-Date Sales, Average Order Value, and Month-over-Month Growth are written as DAX formulas and stored in the model. These measures form the business logic layer that powers all the KPIs and metrics displayed in the final report.

Stage 4: Create Reports and Visualizations

With the data model in place, the analyst moves to the Report View in Power BI Desktop and begins designing the visual layout. Charts, bar graphs, line graphs, pie charts, maps, tables, KPI cards, and slicers are dragged onto report pages and connected to the relevant fields and measures from the data model.

This stage is where raw numbers become visual stories. A well-designed report page allows a business user to immediately understand performance trends, identify outliers, compare categories, and filter results by time period, region, or product without any technical knowledge.

Stage 5: Publish to Power BI Service

Once the report is designed and validated in Power BI Desktop, it is published to the Power BI Service, which is the cloud-based platform where reports are shared and consumed. Publishing uploads both the report layout and the underlying dataset to the cloud workspace.

In the Service, the analyst configures scheduled data refresh so that the dataset automatically updates at defined intervals, such as daily or hourly, keeping the dashboard current without manual intervention. The report is then organized into a workspace and shared with relevant business users or teams through role-based access permissions.

Stage 6: Consume and Collaborate

Business users access the published report through a web browser on the Power BI Service or through the Power BI Mobile application on their devices. They interact with the dashboard by applying filters, drilling down into specific categories, and exploring the data according to their own questions and needs.

At this stage, the workflow delivers its final value. Executives review KPI dashboards before strategic meetings, sales managers track daily performance on their phones, and analysts collaborate on shared workspaces to refine and improve reports based on business feedback.

Question 4: List any four data cleaning tasks that can be performed in Power Query.

Answer:

Power Query is the built-in data transformation engine in Power BI Desktop where raw data is cleaned and prepared before it enters the data model. The following are four essential data cleaning tasks that can be performed directly within Power Query.

1. Removing Duplicate Rows

Power Query allows users to remove duplicate records from a dataset with a single operation. The user selects the relevant column or combination of columns that define uniqueness and applies the Remove Duplicates function. Power Query scans all rows, identifies records that share identical values in the selected fields, retains the first occurrence, and permanently drops all subsequent duplicates from the dataset before it is loaded into the model.

Example: If a Sales table contains Order_ID O101 appearing twice, Remove Duplicates on the Order_ID column retains one clean instance and discards the repeated row.

2. Handling Null and Missing Values

Power Query provides multiple options for dealing with null or empty values in any column. The user can choose to remove rows that contain nulls, replace null values with a defined default such as zero or unknown, or fill values downward or upward from adjacent rows. The chosen action is applied consistently across the entire column during every data refresh.

Example: A null in the Sales_Amount column can be replaced with zero or the row can be removed entirely depending on the business rule defined for that field.

3. Changing and Correcting Data Types

Every column in Power Query is assigned a data type such as text, whole number, decimal, date, or boolean. When data arrives with incorrect types, such as a date stored as plain text or a numeric amount stored as a string, Power Query allows the user to manually reassign the correct data type to each column. This ensures that the data model receives values in the format it expects, preventing type mismatch errors during loading and aggregation.

Example: An Order_Date column imported as text type is changed to Date type so that time-based filtering and trend analysis work correctly in the final report.

4. Splitting and Standardizing Columns

Power Query allows users to split a single column into multiple columns based on a defined delimiter, fix inconsistent formatting, and standardize text values across a column. Operations such as trimming extra spaces, changing text to uppercase or lowercase, replacing specific values, and splitting a full name column into separate first name and last name columns are all available as built-in transformation options.

Example: A Customer_Name column containing values like " rahul mehta" with leading spaces and inconsistent casing can be trimmed and transformed to "Rahul Mehta" using the Trim and Capitalize functions, ensuring consistent and clean name data across all records.

Question 5: Write step by step instructions to load the above dataset.

Answer:

The dataset contains **100 rows** of real estate property data with the following columns: Property_ID, Property_Name, City, Location_Details, Price, Area_sqft, Bedrooms, Bathrooms, Owner_Name, and Contact.

Step 1: Open Power BI Desktop

Launch Power BI Desktop on your computer. You will land on the Home screen. Close the welcome splash screen if it appears. Ensure you are on the **Home tab** in the top ribbon before proceeding.

Step 2: Connect to the Data Source

Click on **Get Data** in the top left of the Home ribbon. A dropdown menu will appear with a list of data source options. Since the file is an Excel workbook with the extension .xlsx, select **Excel Workbook** from the list.

Step 3: Locate and Select the File

A file browser window will open. Navigate to the folder where the file **Real_Estate_Data_100_Rows_csv.xlsx** is saved on your computer. Select the file and click **Open**.

Step 4: Select the Sheet in the Navigator Window

After the file is opened, the **Navigator window** will appear. It displays all the sheets available inside the workbook. You will see one sheet listed which contains the real estate data. Click on the sheet name to preview the data on the right side of the Navigator window. Verify that the preview shows the correct columns including Property_ID, Property_Name, City, Location_Details, Price, Area_sqft, Bedrooms, Bathrooms, Owner_Name, and Contact across 100 rows.

Step 5: Load or Transform the Data

At the bottom of the Navigator window you will see two options.

Select **Transform Data** if you want to open the Power Query Editor first to clean and validate the data before loading. This is the recommended approach since it allows you to check data types, handle any nulls, and verify all columns before the data enters the model.

Select **Load** if the data appears clean in the preview and you want to load it directly into the data model without any transformations.

For this dataset, click **Transform Data** to open Power Query Editor.

Step 6: Verify and Fix Data Types in Power Query Editor

Once inside the Power Query Editor, inspect each column and confirm that the data types are correctly assigned.

Property_ID should be set to **Whole Number** since it is a numeric identifier. Property_Name, City, Location_Details, and Owner_Name should all be set to **Text**. Price and Area_sqft should be set to **Decimal Number** or **Whole Number** since they represent monetary and measurement values. Bedrooms and Bathrooms should be set to **Whole Number**. Contact should be set to **Text** rather than a numeric type since phone numbers should not be used in calculations.

To change a data type, click on the data type icon on the left side of the column header and select the correct type from the dropdown.

Step 7: Rename Columns if Needed

If any column names appear unclear or abbreviated, they can be renamed in Power Query for better readability in reports. Double-click the column header, type the new name, and press Enter. For example, Area_sqft can be renamed to Area in Square Feet for better clarity on dashboards.

Step 8: Check for Nulls and Duplicates

In Power Query Editor, scan each column for null or empty values by checking the column quality bar visible below each column header. It shows the percentage of valid, empty, and error values. If any nulls are found in critical columns such as Price or Property_ID, handle them by either removing those rows or replacing the nulls with appropriate default values before loading.

Also verify that Property_ID contains no duplicates by right-clicking the column and selecting Remove Duplicates to ensure each property has a unique identifier.

Step 9: Close and Apply

Once all data type corrections, null handling, and column validations are complete, click **Close and Apply** in the top left corner of the Power Query Editor. This applies all transformation steps and loads the cleaned data into the Power BI data model. A loading progress bar will appear briefly while the data is being processed.

Step 10: Verify the Loaded Data in Power BI Desktop

After loading is complete, navigate to the **Data View** by clicking the table icon on the left sidebar of Power BI Desktop. Confirm that all 100 rows are present and all 10 columns are correctly loaded with the right values and data types. The dataset is now ready for building visualizations, creating DAX measures, and designing reports and dashboards.

Question 6: Define Data View, Report View, and Model View. Explain the purpose of each view.

Answer:

Power BI Desktop provides three distinct views that together cover the complete process of working with data, from inspecting raw values to building relationships and designing final reports. Each view serves a specific and non-overlapping purpose in the analytics workflow.

1. Data View

Data View is the section of Power BI Desktop that displays the actual data loaded into the model in a tabular row-and-column format, similar to how data appears in an Excel spreadsheet. It is accessed by clicking the table icon on the left sidebar of Power BI Desktop.

Purpose: Data View is used to inspect, verify, and interact with the raw data that has been loaded into the model. After connecting to a data source and completing transformations in Power Query, an analyst uses Data View to confirm that all rows and columns have been loaded correctly, that data types are accurate, and that values appear as expected. It is also the place where calculated columns are created using DAX formulas. A calculated column adds a new column to an existing table by computing a value row by row based on other columns in the same table. For example, a new column called Price per Sqft can be created by dividing the Price column by the Area_sqft column directly in Data View. This view does not allow visual design or relationship building. Its sole purpose is data inspection and calculated column creation.

In the context of the Real Estate dataset: An analyst would open Data View after loading the dataset to verify that all 100 property records are present, that Price is stored as a number and not text, that City values are consistent, and that no unexpected nulls appear in critical columns.

2. Report View

Report View is the primary design canvas in Power BI Desktop where analysts and developers build the visual reports and dashboards that business users will ultimately consume. It is the default view that opens when Power BI Desktop is launched and is accessed by clicking the bar chart icon on the left sidebar.

Purpose: Report View is used to design and arrange all visual elements of a report including charts, graphs, tables, KPI cards, maps, slicers, and text boxes onto one or more report pages. The analyst drags fields from the data model into the visual properties panel on the right side to bind data to visuals. Multiple report pages can be created within a single file, each focusing on a different aspect of the data such as sales performance, regional analysis, or customer

segmentation. Report View is also where slicers and filters are added to make the report interactive, allowing business users to dynamically filter the data displayed on the dashboard by selecting values such as a specific city, date range, or product category. No data editing or relationship building happens here. This view is purely focused on storytelling through visualization.

In the context of the Real Estate dataset: An analyst would use Report View to build a dashboard showing average property price by city, a bar chart comparing number of bedrooms across properties, a map visual plotting properties by location, and a slicer allowing users to filter all visuals by city or number of bedrooms.

3. Model View

Model View is the section of Power BI Desktop that displays the data model as a visual diagram showing all loaded tables and the relationships defined between them. It is accessed by clicking the relationship diagram icon on the left sidebar.

Purpose: Model View is used to define, manage, and visualize the relationships between different tables in the data model. When data is loaded from multiple tables, Power BI needs to understand how those tables are connected to each other so that filters and aggregations work correctly across tables. In Model View, the analyst draws relationship lines between tables by connecting matching fields, for example linking a Sales table to a Customers table through a shared Customer_ID field. The analyst also configures the cardinality of each relationship, meaning whether it is a one-to-many, many-to-one, or many-to-many relationship, and sets the cross-filter direction. This view also allows analysts to hide unnecessary columns from the report canvas, organize tables visually for clarity, and review the overall structure of the data model. A well-designed model in this view is what makes DAX measures, cross-table filtering, and multi-table reports work correctly in Report View.

In the context of the Real Estate dataset: If the dataset were expanded to include a separate Owners table and a Cities table, Model View would be where the analyst connects the Real Estate table to the Owners table through Owner_Name and to the Cities table through City, enabling cross-table filtering and unified reporting across all three tables.

Question 7: Discuss the different data sources that Power BI supports.

Answer:

Power BI is designed to connect to a wide variety of data sources, making it one of the most flexible business intelligence tools available. Rather than being limited to a single type of data, Power BI allows analysts to pull data from files, databases, cloud platforms, online services, and even web pages, all within the same tool. Understanding the categories of supported data sources is important because the choice of source directly affects how data is extracted, refreshed, and used in reports.

1. File-Based Data Sources

File-based sources are the most commonly used starting point for analysts working with locally stored data. Power BI supports direct connection to the following file types.

Excel Workbooks (.xlsx, .xls) are the most widely used file source. Power BI can connect to Excel files and import data from individual sheets or named tables within a workbook. Since Excel is the most common data storage tool in business environments, this connection type makes Power BI accessible to teams that already manage their data in spreadsheets.

CSV and Text Files (.csv, .txt) are flat file formats where values are separated by commas or other delimiters. Power BI can read these files directly and automatically detects column separators and data types during import.

PDF Files can also be connected as a data source. Power BI extracts tables embedded within PDF documents, which is useful when reports or official data are only available in PDF format.

XML and JSON Files are structured file formats commonly used in data exports from web services and APIs. Power BI supports direct import of both formats and can navigate nested structures within JSON to extract the relevant data fields.

In the context of the Real Estate dataset: The file `Real_Estate_Data_100_Rows_csv.xlsx` is an example of an Excel file-based data source connected directly to Power BI Desktop.

2. Relational Database Sources

Power BI connects directly to relational database management systems, allowing analysts to query structured data stored in enterprise-grade databases. These connections support both import mode and live query mode.

Microsoft SQL Server is the most natively supported database in Power BI due to both being Microsoft products. Power BI can connect to SQL Server instances running locally or on a network and can execute custom SQL queries during the connection.

MySQL is an open-source relational database widely used in web applications and mid-sized businesses. Power BI connects to MySQL using a connector that requires the MySQL ODBC driver to be installed.

PostgreSQL is another open-source database popular in data engineering environments. Power BI supports PostgreSQL connections with similar configuration requirements.

Oracle Database is an enterprise database system used by large corporations. Power BI supports Oracle connections but requires the Oracle client software to be installed on the same machine running Power BI Desktop.

IBM DB2 and **Sybase** are also supported for organizations running legacy enterprise database systems.

3. Cloud-Based Data Sources

As organizations increasingly store data on cloud platforms, Power BI provides native connectors to the most widely used cloud data services.

Microsoft Azure offers the deepest integration with Power BI since both are Microsoft products. Power BI connects directly to Azure SQL Database, Azure Synapse Analytics, Azure Data Lake Storage, Azure Blob Storage, and Azure Cosmos DB, making it the preferred analytics layer for organizations using the Azure ecosystem.

Amazon Redshift is the cloud data warehouse offered by Amazon Web Services. Power BI supports direct connection to Redshift, enabling organizations using AWS infrastructure to build Power BI reports on top of their cloud data.

Google BigQuery is Google's cloud data warehouse. Power BI provides a native connector to BigQuery, allowing analysts to query large datasets stored in Google Cloud directly from Power BI Desktop.

Snowflake is a cloud-native data warehouse platform that has become widely adopted across industries. Power BI connects to Snowflake and supports both import and DirectQuery modes, making it a popular combination for modern data stack implementations.

4. Online Services and SaaS Platforms

Power BI connects to a range of software-as-a-service platforms that businesses use daily, allowing operational data from these tools to flow directly into reports without manual data exports.

Microsoft SharePoint allows Power BI to connect to lists and libraries stored in SharePoint Online, which is commonly used for project tracking and document management in organizations using Microsoft 365.

Microsoft Dynamics 365 is Microsoft's CRM and ERP platform. Power BI has deep native integration with Dynamics 365, enabling sales, finance, and operations reports to be built directly from live CRM data.

Salesforce is the most widely used CRM platform globally. Power BI provides a native Salesforce connector that allows connection to Salesforce objects such as leads, opportunities, accounts, and cases.

Google Analytics is supported through a native connector that pulls web traffic, user behavior, and conversion data directly into Power BI, enabling marketing teams to build web performance dashboards.

Microsoft Exchange and Outlook data including email and calendar information can also be connected for productivity and communication analytics.

5. Web and API-Based Sources

Power BI can retrieve data from publicly available web pages and REST APIs, making it possible to incorporate external data into reports without waiting for database exports.

Web Connector allows Power BI to connect to a URL and extract tables found on that web page. This is useful for pulling data from publicly available sources such as government statistics pages, financial data portals, or Wikipedia tables.

REST APIs can be connected through the Web connector by providing the API endpoint URL along with any required authentication headers. Power BI can consume JSON or XML responses from APIs and transform them into tabular data for reporting.

OData Feeds are a standardized protocol for exposing data through APIs. Power BI has a dedicated OData connector that simplifies connection to services that expose their data in OData format.

6. Big Data and Analytics Platforms

For organizations working with large-scale data infrastructure, Power BI connects to big data and analytical processing platforms.

Apache Spark and **Azure HDInsight** are supported for organizations processing large datasets through distributed computing frameworks. Power BI can connect to Spark clusters and query data that has been processed at scale.

Hadoop File System (HDFS) is supported for organizations storing data in Hadoop-based data lakes, allowing Power BI to retrieve processed data from distributed storage.

Databricks is a unified analytics platform built on Apache Spark. Power BI provides a native connector to Databricks, making it a common pairing in modern data engineering workflows.

7. Other Sources

Microsoft Access databases, which are commonly used in small businesses and legacy systems, are fully supported by Power BI through a direct connector.

ODBC and OLE DB connections allow Power BI to connect to any data source that exposes a standard ODBC or OLE DB interface, effectively making any database with a compatible driver accessible to Power BI even if no native connector exists.

Blank Query allows analysts to write custom M language queries in Power Query to connect to unconventional sources or construct data manually when no standard connector applies.

Question 8: Split Owner Name to create two new columns as First Name and Last Name

Answer:

Split Owner Name into First Name and Last Name

This task is performed in **Power Query Editor** in Power BI Desktop. The Owner_Name column contains full names such as "Rahul Mehta" and "Anita Sharma" which need to be split into two separate columns.

Using Power BI Power Query (Step-by-Step)

Step 1: Open Power Query Editor In Power BI Desktop, click on Transform Data from the Home ribbon. This opens the Power Query Editor where all data transformation tasks are performed.

Step 2: Select the Owner_Name Column In the Power Query Editor, locate and click on the Owner_Name column header to select it. The entire column will be highlighted in green.

Step 3: Split the Column by Delimiter With the Owner_Name column selected, go to the Home tab or Transform tab in the Power Query ribbon. Click on Split Column and from the dropdown select By Delimiter.

Step 4: Configure the Split Settings A dialog box will appear. In the delimiter field, select Space from the dropdown since first name and last name are separated by a space. Under the option for where to split, select Left-most Delimiter so that the column splits only at the first space, correctly separating the first name from the last name even if the last name contains multiple words. Click OK.

Step 5: Rename the New Columns Power Query will automatically create two new columns named Owner_Name.1 and Owner_Name.2. Double-click on Owner_Name.1 and rename it to First_Name. Double-click on Owner_Name.2 and rename it to Last_Name.

Step 6: Verify the Split Scroll through the newly created columns to confirm that the split has been applied correctly across all 100 rows. Rahul Mehta should now show Rahul in First_Name and Mehta in Last_Name.

Step 7: Close and Apply Once verified, click Close and Apply in the top left of Power Query Editor to apply the transformation and reload the data into the Power BI data model.

Expected Output After Splitting

Property_ID	Property_Name	Owner_Name	First_Name	Last_Name
101	Green Heights	Rahul Mehta	Rahul	Mehta
102	Sunrise Apartments	Anita Sharma	Anita	Sharma
103	Lake View Residency	Ramesh Kumar	Ramesh	Kumar

Why This is Useful in Power BI Reports

Once the Owner_Name is split into First_Name and Last_Name, reports become more flexible. Dashboards can display owner names in a formatted way such as Last Name followed by First Name. Filters and slicers can be applied independently on first name or last name. Sorting by last name alphabetically becomes straightforward, which is the standard convention in professional property listings and owner directories. The original Owner_Name column can be retained or removed depending on whether the combined form is still needed elsewhere in the report.